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Code No: 138GW

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
RENEWABLE ENERGY SOURCES
(Common to CE, EEE, ME, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Explain the primary and secondary energy resources with examples?	[2]
b)	Differentiate between Conventional resources and Non-conventional sources.	[3]
c)	Give the different applications of solar photovoltaic cells.	[2]
d)	What are thermal conversion devices of solar energy system?	[3]
e)	Why do wind energy potential is low in India?	[2]
f)	Comment on the future availability trend of hybrid systems with wind.	[3]
g)	Classify the biomass resources?	[2]
h)	What are the applications of biogas plants?	[3]
i)	Explain the principle of geothermal energy.	[2]
j)	Write about the hot springs and steam ejection.	[3]

PART – B

(50 Marks)

2.a)	Explain the role and potential of new and renewable energy sources in global scenario.	
b)	Other than price, what other factors influence the acceptance of renewable energy supply over traditional energy?	[5+5]
	OR	
3.a)	Explain briefly the limitations and advantages of the hybrid systems.	
b)	How can you understand the concept of Global climate change and its impact?	[5+5]
4.a)	How can solar thermal energy be measured and stored?	
b)	Explain briefly the solar thermal storage devices.	[5+5]
	OR	
5.a)	Write a short notes on solar radiation measurement and estimation.	
b)	How do you calculate solar radiation on tilted surfaces and its advantages?	[5+5]

6.a) Explain the factors influencing the wind mill design.
b) Explain the status of wind energy potential and installation in India. [5+5]

OR

7.a) Explain the advantages and limitations of wind energy banking concept.
b) Describe with a neat sketch the working of a wind energy system with design steps.[5+5]

8.a) What are the prospects of biomass gasification energy?
b) Compare and contrast the pyrolysis and liquefaction. [5+5]

OR

9.a) Briefly give the design and constructional features of bio energy system.
b) Explain the Biogas energy program in India with numerical data. [5+5]

10.a) Explain how the locations may have influence on enhanced power output from a tidal-current power device.
b) Explain the scope of and development OTEC. [5+5]

OR

11.a) Briefly discuss vapor dominated geothermal plant with a diagram.
b) Explain the basic components of Small hydroelectric scheme with a layout arrangement and its working. [5+5]

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